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Thermal Comfort in the Tropical South East Asia Region

Tri H. Karyono*

The current ISO and ASHRAE standards for thermal comfort, recommend universal values for comfort temperatures. They do not distinguish between people living in temperate climates such as Europe and those in warm-humid climates such as the South East Asia region.

This paper tries to assess whether the current standards on thermal comfort, which were mainly established from research in mild climates would be fit for people in the warm-humid South East Asia region.

Glossary

Comfort range (comfort temperature): A range of temperature which is defined between 'slightly cool' (-1) and 'slightly warm' (+1) on subjects' comfort votes. It was proposed by Fanger [1] that within the predicted mean vote (PMV) of -1 and +1 around 75% of subjects would be satisfied with their thermal environment [1]

ET : effective temperature: A temperature which is measured as a combination effect of dry bulb (air) temperature, humidity, and air velocity.

ET* : corrected effective temperature: A temperature which is measured as a combination effect of air temperature, mean radiant temperature, humidity, and air velocity.

Neutral temperature: A temperature in which the highest percentage of a group of people being satisfied with their thermal environment would be reached. It is stated by Fanger [1] that in this temperature around 95% of subjects would be satisfied with their thermal environment.

RH : Relative humidity:

T_a : Air Temperature

T_o : Operative temperature: A temperature which is calculated as the average of air and mean radiant temperatures.

T_{eq} : Equivalent Temperature: A temperature which is measured as a combination effect of air temperature, mean radiant temperature, and air velocity.

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1. Introduction

Politically, the South East Asia countries consists of Indonesia, Malaysia, Singapore, Brunei, Thailand and The Philippines. However, the country of Papua New Guinea which is on the east border of Indonesia and is situated on the tropical area is also within this region, and therefore will also be discussed. A number of thermal comfort studies have been done in this region: in Indonesia, by Mom and Wiesebron between 1936 and 1940 in the city of Bandung [2,3], and by Karyono in Jakarta in 1993 [4]; in Singapore, by Webb in 1950 [5], Ellis in 1952 and 1953 [6,7], and de Dear in 1990 [8,9,10]; in Bangkok, Thailand, by Busch in 1988 [11]; in Port Moresby, New Guinea [12,13], by Ballantyne in 1967 and 1979. This paper reviews these studies and compares the results with the current ISO [14] and ASHRAE [15] standards for thermal comfort.

2. Review of Thermal Comfort Studies in the Tropical South East Asia Region

2.1. Indonesia

Jakarta is situated on the 6° south latitude, possesses a warm-humid tropical climate with little temperature variation throughout the year. The minimum average temperature is 23°C (midnight towards morning) and the maximum average is 33°C (midday). The average relative humidity is between 69% and 90%, and the average air velocity is between 0.2 and 0.8 m/s.

At least two studies of thermal comfort have been carried out in Indonesia, the first by Mom and Wiesebron [2,3] and the second by Karyono in the capital city of Jakarta in 1993 [4].

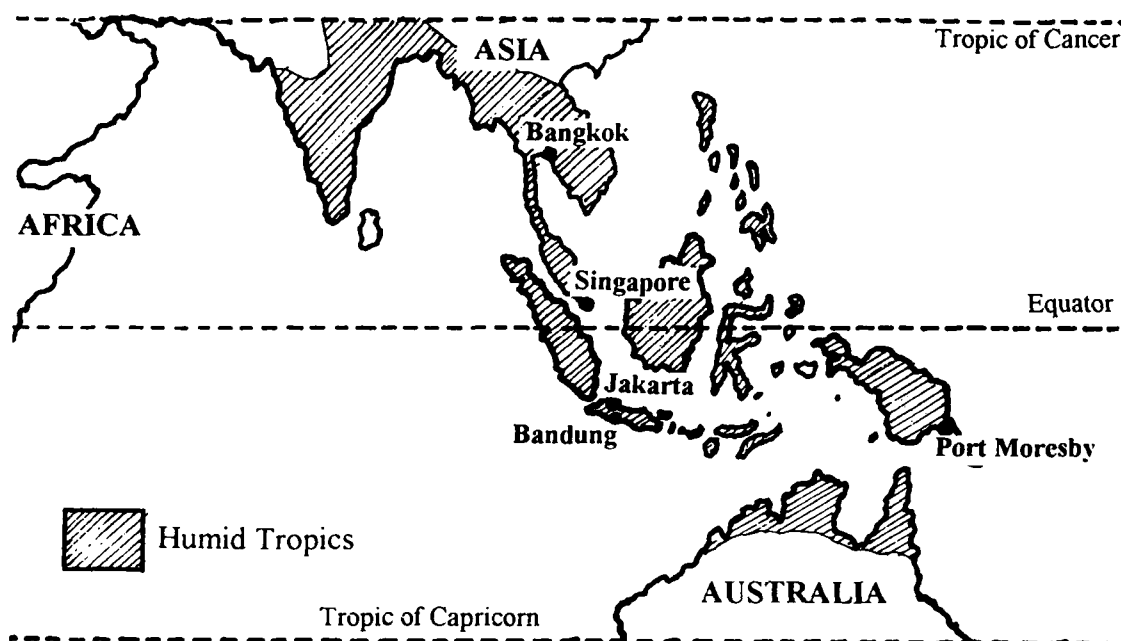


Figure 1 Geography of the South East Asian Region

Mom and Wiesebron conducted a study in Bandung in West Java between 1936 and 1940 [2,3]. This study was carried out in the Technische Hoogeschool Bandung (currently the Bandung Institute of Technology, ITB). There were three different groups of subjects: Europeans who lived in Bandung, Chinese who lived in Bandung, and Indonesians. The last group was divided into two sub-groups: educated and uneducated subjects. There is no information concerning the number in each group. The Indonesian wore tropical (thin) clothes, the other subjects wore suits.

A five-point scale of thermal comfort vote was used: uncomfortable cold, comfortable cool, comfort or optimum comfort, comfortable warm and uncomfortable hot. The air velocities during the measurement were taken to be between 0.1m/s and 0.2m/s. The subjects' overall 'optimum comfort' temperature was between 19.3° and 25.1°C T_{eq} (effective temperature). The 19.3°C T_{eq} was equal to 22.5°C T_a (air temperature) with RH (relative humidity) of 35% or, 21°C T_a with 90% RH; while the 25.1°C T_{eq} was equal to 29°C T_a with 30% RH, or, 26°C T_a with 90% RH.

The result from the Indonesian subjects alone was 22.8°C T_{eq} (or 24°C T_a with 80%RH) to 25.8°C T_{eq} (or 28°C T_a with 70%RH) for 'optimum comfort'. The median of this range, the neutral temperature, had a value of 24.4°C T_{eq} or roughly 26°C T_a . The lower limit of 'comfortable cool' was 20.5°C T_{eq} (or 23°C T_a with 50% RH), while the upper limit of 'comfortable warm' was 27.1°C T_{eq} (or 31°C T_a with 60% RH). Therefore the 'comfort range' of the Indonesian subjects who lived in Bandung was 20.5°C T_{eq} (or 23°C T_a with 50% RH) to 27.1°C T_{eq} (or 31°C T_a with 60% RH).

The other thermal comfort study was done in Indonesia by Karyono in 1993 in the capital city of Jakarta. This field study involved 596 Indonesian office workers from seven different office buildings, one naturally ventilated, one hybrid and five air conditioned [4].

Workers were asked to give their thermal sensation votes, based on a seven-point thermal sensation scale: cold (-3), cool (-2), slightly cool (-1), neutral (0), slightly warm (+1), warm (+2), and hot (+3), while at the same time indoor climatic parameters of air temperature (T_a), operative

temperature (T_o), equivalent temperature (T_{eq}) and relative humidity (RH) were measured.

An effort was made to exclude subjects receiving asymmetric thermal radiation as such as direct solar radiation on parts of their bodies, and those is excessive air flow (draught) from open windows or diffusers. All the measurements were carried out between 10 a.m. and 4 p.m.

The indoor air temperatures and relative humidity were measured with a thermo-hygrometer, while the operative temperatures and equivalent temperatures were measured by using Thermal Comfort Meter type 1212, a portable instrument used for measuring the effect of thermal environment on human comfort by stimulating the thermal balance of the human body [16]. This instrument is battery-powered, having a dimension of 13.3 cm in height, 2.10 cm in width, and 20.0 cm in depth, which senses environmental condition through the Comfort Transducer - a grey microphone's look-like. The transducer is electronic in its operation and is connected by a cable.

The reading of the measurement is displayed on liquid crystal digital on the front panel of the instrument. Three parameters, clothing value, activity level, and vapour pressure, must be pre-set before the measurement is taken.

The Thermal Comfort Meter is designed to be operated in environments where the temperature does not fall outside the range of + 5° to + 40° C and the humidity under or equal to 90%, and is powered by six 'D size' non-rechargeable alkaline cells of 1.5 volt each. Before being used in the actual measurements, this instrument had been checked for calibration.

The 596 subjects were office workers, 345 males and 227 females indicated their gender (24 subjects missed to indicate it in the questionnaire). Subjects were between 19 and 53 years of age having a mean of 32.6 years with standard deviation of 7.4 years. In height, the shortest subjects was 145 cm while the tallest was 183 cm, with a mean of 162.58 cm and standard deviation of 7.14 cm. The lightest subject was 38 kg and the heaviest was 100 kg, with an average of 56.19 kg and standard deviation

of 10.12 kg. The range of subjects' Dubois area was between 1.28 to 2.11 m², with a mean of 1.6 m² and standard deviation of 0.16 m². The origin of subjects were various although they were all Indonesian citizens. They came from more than ten ethnic groups in which the majority, 227 were Javanese, while the others varying in numbers came from other Indonesian islands, such as Sumatra, Kalimantan (Southern part of Borneo), Sulawesi, and Bali.

The activity level of all subjects involved in this measurement was taken as 1 met (50 kcal/h.m²) since the subjects were detected to have been seated for about 15 minutes. This is in agreement with the value recommended for the light office activity [14,16]. The clothing value of subjects during measurements were estimated to be 0.6 clo (93% subjects), 0.8 clo (6.7% subjects) and some 0.3% were 1.0 clo and 1.2 clo. While the vapour pressure were calculated to have a range of between 1.8 kPa and 3.0 kPa with interval of 0.3 kPa.

This study showed that subjects' neutral temperature was 26.4°C T_{n_2} or 26.7°C T_{n_0} (operative temperature) or 25.3°C T_{eq} (equivalent temperature). This neutral temperature is about 2.5°C higher than those recommended by either ISO or ASHRAE. No clear difference was found in the neutral temperatures of subjects in naturally ventilated and air conditioned buildings.

To compare the results of Mom and Wiesebron's study to that of Karyono, it can be seen that the neutral temperature of people in Jakarta was somewhat higher than those for people in Bandung (24.4°C ET or 26°C T_n). From above figures it seems that people who live in the city of Bandung with a range of outdoor temperatures of 18°C to 28°C, which is lower than the average for Jakarta (23°C to 33°C) also preferred lower comfort temperature. The observation that the people in Bandung, which has lower average outdoor temperatures than Jakarta, preferred a lower neutral temperature could be an indication that human adaptation to the prevailing climate does exist, and that it affects comfort temperatures of people who live in cities with different average outdoor temperatures.

2.2. Singapore

Singapore is located in the latitude of 1°20', and has a temperature variation similar to that of Jakarta, that is around 23°C average minimum and 33°C of average maximum.

In 1950, Webb [5] carried out a thermal comfort study in Singapore, involving 16 residents of Singapore, this included two Europeans who had lived for more than twenty years in Singapore. Observations were made about every half hour, using a wet and dry bulb whirling thermometer, a kata-thermometer and a globe thermometer. A nine-point scale of thermal sensation was used to indicate the subjects' thermal response to the surrounding thermal environment. It was found in this study that the subjects' comfort temperature was 26.2°C (79°F) ET.

Another thermal comfort study has been done in Singapore by Ellis in 1952 and 1953. The first study, in 1952 [6], was carried out in the aircraft-repair ship on the passage from Singapore to Hong Kong and a double passage between Singapore and Japan. A total of 5211 comfort votes were collected from subjects of the ship's crews and seamen. Ellis found from this study that the upper limit of subjects comfort temperature was 26.1°C (79°F) ET or 30.0°C (86°F) dry bulb temperature (or air temperature, T_a). The second study, in 1953 [7], was carried out in a university college, a health office and homes, and involved 118 subjects of European, Chinese, Indian, Ceylonese, Malay and Eurasian. Comfort temperature for the Europeans was found to be between 22 and 25.5°C ET or 24.5 and 27.8°C T_n .

In 1990, Richard de Dear [8] undertook a comfort study in a thermal chamber in Singapore, involving 32 college-age subjects consisted of 16 males and 16 females. The preferred temperature was found to be 25.4°C T_a , while the preferred mean skin temperature was 33.1°C. A further study was undertaken by de Dear [9] in the same place involving 98 college-age subjects, and he found that the upper limit of the acceptable comfort zone was 27.6°C at 70% RH and 27.9°C at 35% RH.

In 1990 de Dear carried out another comfort study in Singapore [10]. This study involved 583 respondents from naturally ventilated dwellings and 235 respondents from air-conditioned offices buildings. Four climatic parameters, air temperature, mean radiant temperature, humidity and air velocity were measured whilst subjects were asked to give comfort votes based on the standard seven-point scale (hot=+3; warm=+2; slightly warm=+1; neutral=0; slightly cool=-1; cool=-2; cold=-3). The preferred comfort sensation of McIntyre [17] (does the room feel cool, warm, or neutral/just right at the moment?) was also asked to the respondents.

Subjects' thermal comfort (the neutrality) in naturally ventilated dwellings was found to be 28.5°C T_{n_0} , whilst in air conditioned offices was 24.2°C T_{n_0} . This is based on the mean clothing insulation value of 0.26 clo and a mean of metabolic rate of 70 W/m² of subjects in naturally ventilated buildings, and 0.44 clo of subjects' clothing insulation value and 67 W/m² of subjects' metabolic rate in air conditioned buildings.

2.3. Thailand

A thermal comfort study was carried out by Busch [11] during the hot season (in April) and the wet season (in July) in 1988 in the capital city of Bangkok, Thailand. The prevailing outdoor temperature of Bangkok (which was not mentioned in his report) is about the same as that in Jakarta, or slightly higher, which is about 24°C average minimum and 34°C average maximum.

A total of 1146 office workers from four different office buildings (two naturally ventilated and two air conditioned) participated in this study, 669 women and 476 men. The physical environmental parameters measured were dry-bulb temperature, globe temperature, relative humidity and air velocity. The subjects' thermal sensation was recorded on the seven-point ASHRAE scale (cold, cool, slightly cool, neutral, slightly warm, warm, hot) [18] and three-point McIntyre scale (warmer, no change, cooler).

This study showed that the average neutral temperature of the subjects was 25.0°C ET*. The neutral temperature of subjects in the wet season was 25.7°C ET*, which was 1.2°C ET* higher than that during the hot season. When separate analyses were made of naturally ventilated and air-conditioned buildings, subjects in the naturally ventilated buildings preferred 28.5°C ET*, while subjects in the air-conditioned buildings preferred 24.5°C ET* for their neutral temperatures. It was also found that female subjects preferred 0.8°C ET* higher than their male colleagues.

2.4. Port Moresby, New Guinea

Ballantyne [12] undertook a thermal comfort study in Port Moresby, New Guinea (about 10° South latitude), which was reported in 1967. This study involved 32 expatriate Caucasian subjects. The field study was carried out in their houses, using a seven-point scale of comfort vote (much too cool=1; too cool=2; cool=3; neutral=4; warm=5; too warm=6; much too warm=7) to weigh the subjects' thermal sensations. Three climatic parameters, dry bulb (T_a), relative humidity (RH), and wind speed (V_a), were measured during the study. Subjects' activity levels and clothing insulation value were not weighted.

Table 1.
Neutral Temperatures and Comfort Ranges of Subjects in Tropical South East Asia

Study by (first author)	Time	Type of Study	Country/ Regions	Ethnic	No. of Subjects	Comfort (°C)
Mom	1937	Thermal Chamber	Bandung, Indonesia	Indonesian European	20	26.0 Ta
Webb	1950	Field Study	Singapore	Malay European	16	26.2 ET
Ellis	1952	Field study	Journey between Singapore-Hong Kong	no data	5211 votes	26.1 ET (30.0 Ta)
Ellis	1953	Field study	Singapore	Eurasian. Chinese. Indian. Ceylonese. Malay	118	22-25.5 ET (24.5-27.8 Ta)
Ballantyne	1967	Field Study	Port Moresby	Caucasian	34	25.6 Ta
Ballantyne	1979	Thermal Chamber	Port Moresby	Melanesian Caucasian	64 28	26.7 Ta 25.0 Ta
Busch	1988	Field Study	Bangkok, Thailand	Thai	1100	28.5 ET* (NV) 24.5 ET* (AC) 25.4 ET* (all)
De Dear	1990	Thermal Chamber	Singapore 1	Singaporean	32	25.4 Ta
De Dear	1990	Thermal Chamber	Singapore 2	Singaporean	98	upper limit: 27.6Ta (70% RH) 27.9Ta (35%RH)
De Dear	1990	Field study	Singapore 3	Singaporean	583 235	28.5 To (NV) 24.2 To (AC)
Karyono	1993	Field Study	Jakarta, Indonesia	Indonesian	596	26.4 Ta 26.7 To 25.3 Teq

Note:

In the still air ($V_a=0$), and mean radiant temperature (T_{mrt}) = air temperature (T_a). the value of $T_a = T_o = T_{eq}$. The higher the T_{mrt} than the T_a , the higher the value of T_o than T_a and T_{eq} . The greater the air velocity, the lower the value of T_{eq} than T_a and T_o

The results of this study showed that subjects' neutral temperature was found to be 25.6°C T_z .

In 1979 Ballantyne [13] reported a similar study in the same city. The 92 subjects consisted of 34 Melanesians' male and 30 females and 14 Caucasian males and 14 females. Air temperature, water vapour pressure, and wind speed were measured. A five-point scale of comfort vote (cold, cool, just right, warm, hot) was used to measure subjects thermal sensation. Activity and clothing values were not weighted during the study.

From this study, it was found that the neutral temperature for Melanesians was 26.7°C T_z (1.4°C higher than the prediction by Fanger's comfort equation) and 25.0°C T_z for Caucasians subjects. The neutral temperature of the Melanesians seems to be similar to the Indonesian subjects in the Jakarta study (26.4°C T_z).

3. Discussion

Table 1 shows that people in the South East Asia region under the similar outdoor temperatures, preferred quite similar comfort or neutral temperatures. The only exceptional is in Ellis study in 1953, which found that the subjects preferred a slightly lower comfort temperature.

The comfort temperatures of the South East Asian subjects were above 24°C in any terms of temperatures, and they must be even higher when converted into operative temperature (T_o), since the mean radiant temperature usually higher than the indoor air temperature. Compared to both ASHRAE and ISO standards [14,15] which recommend comfort temperatures for people live in such climatic condition about 24°C $T_o \pm 1^\circ$, the actual requirement of comfort temperatures of people from the South East Asian region were found to be higher. This has an implication for the building energy consumption, particularly when the building is air conditioned, since setting up higher indoor temperatures would reduce energy for air cooling in the building.

On average, the Caucasians (Europeans or Australians) preferred temperature slightly lower than those of indigenous tropical men, such as the Indonesians, Melanesians, Thais, Singaporeans and Malays.

Although results of Karyono's study in Jakarta showed no difference on neutral temperature between subjects in air-conditioned and naturally ventilated buildings, Busch [11] and de Dear [10] showed that subjects in naturally ventilated buildings preferred higher comfort temperatures than those in air conditioned ones. However, in the de Dear study subjects in naturally ventilated buildings (dwellings) wore only 0.26 clothing insulation value compared with those in air conditioned buildings (offices) who wore 0.44 clo clothing.

4. Conclusions

The various studies show that people living in the warm humid tropical regions of South East Asia prefer similar neutral temperatures, that is, around 25° to 30°C. These figures are 1° to 6°C higher than those recommended by the ISO and ASHRAE standards. The preferred higher temperatures may be the result of adaptation by people in South East Asia to higher outdoor temperatures, a factor not recognised in the current standard.

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